

VARUN DOBHAL

B.Tech Computer Science Engineering (Cybersecurity)

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Summary

Backend-focused Software Developer with experience in scalable applications, secure systems, and blockchain-based solutions. Published IEEE researcher and Indian patent holder focused on cybersecurity, blockchain security, applied cryptography, and quantum computing.

Education

Graphic Era Deemed University

2023 – 2027

B.Tech in Computer Science Engineering (Cybersecurity)

CGPA: 7.9

Skills

- **Languages:** JavaScript (ES6+), Rust, Python, C, Solidity
- **Frameworks:** React.js, Node.js, Express.js, Tauri, Django, Rocket
- **Databases:** PostgreSQL, MongoDB, Redis, SQLite, Supabase
- **Tools:** Docker, Git, Postman, Linux (Kali/Ubuntu Server), Mockoon
- **Core CS:** Data Structures & Algorithms, Operating Systems, Compiler Design, DBMS, Computer Networks

Projects

Independent ERP System for Graphic Era University (Ongoing)

Jan 2026 – Present

- Engineering a fully in-house ERP platform and production-grade server infrastructure to support 40K–45K students, reducing reliance on third-party solutions by 100%
- Building scalable student, attendance, and administrative modules capable of processing 5,000+ concurrent requests with optimized database indexing and query performance enhancements.

CTF Platform (National-Level Hackathon)

Feb 2026 – Apr 2026

- Developed a full-stack CTF platform supporting 100+ teams with a real-time WebSocket-based leaderboard, achieving sub-500ms latency for score updates
- Implemented automated flag validation, challenge administration, and proctoring mechanisms to detect and prevent unfair practices during competitions

QR-Based Event Entry Verification System

Jan 2026 – Mar 2026

- Built a high-traffic verification platform for 4,500+ attendees, ensuring zero duplicate entries through a PostgreSQL-backed audit system
- Automated Excel-to-database ingestion and QR generation, reducing the event check-in time per attendee by approximately 60% compared to manual logs

Adaptive Computational Hardness for Quantum-Aware Systems (Research)

Mar 2026 – Present

- Architecting a quantum-aware cryptographic framework utilizing adaptive computational hardness to mitigate Shor's algorithm-based threats
- Developing dynamic cost-escalation mechanisms that increase attack complexity by a factor of 10x against brute-force and quantum-accelerated attempts
- Conducting simulation and benchmarking studies to evaluate security-performance trade-offs in post-quantum environments
- **Authors:** V. Dobhal, M. Wazid

Publications

- V.Dobhal, S.Mittal, M.Wazid, S.Saha, A.K.Das, S.Pal, J.J.P.C.Rodrigues, "Blockchain-Based Secure Product Authenticity Verification for Industrial Networks," IEEE ICBC 2026, Brisbane, Australia

Patents

- Indian Patent (Published, 2026) — Secure Data Erasure System for Enterprise Storage (202611016955)
- Indian Patent (Published, 2024) — Blockchain-Based Product Authenticity System (202411082477)